Seat: TBD PETER ANTEATER PANTEATE

CompSci 161 Diagnostic Exam 2-1, Fall 2025

DO NOT OPEN EXAM UNTIL INSTRUCTED TO DO SO

SILENCE AND STORE ALL ELECTRONICS

This is a diagnostic exam intended to help you evaluate your readiness for the real exam.

The following rules apply to you, whether you think they do or not. Read and understand them; failure to abide by these rules, or directions given by course staff during the exam, may result in disciplinary action, including but not limited to a failing grade in the class.

- This exam is solely for students enrolled in this lecture. Anyone not enrolled in this lecture may not take an exam.
- You may not open the exam or begin writing until the instructor has explicitly given you permission to do so.
- Keep acceptable identification readily accessible during the test. Proctors may request to see it.
- This exam is closed book, closed notes, and is individual effort. Once course staff begin passing out exams, you may not communicate with anyone other than proctors for any reason, nor may you have electronics, including calculators watches and phones, available to you during the test for any reason. **YOU DO NOT NEED A CALCULATOR!**
- If you leave your seat during the test for any reason, your instructor may collect it and deem you to have turned it in. Do not ask proctors for an exemption to this, they are not authorized to grant such.
- You must take the exam in your assigned seat unless the professor (not a TA) tells you otherwise. The instructor will call to cease writing at 6:45 PM, at which point you must immediately cease writing and close the exam. You may not write any further at that point, including finishing one's current sentence.
- If you believe a question is ambiguous, write at least two reasonable interpretations and indicate clearly which one you will be using. Then answer your question with that assumption. Unless your interpretation makes the problem much more trivial than intended, we will grade your response as if one of us had made that clarification.
- The purpose of the real exam is to evaluate how well *you* understand the material presented in the course. It is an academic integrity violation to do anything that subverts the goals of this assessment including, but not limited to, not doing your own work or submitting that of anyone else.
- We will only grade responses marked in the space provided for each question.

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1. (1.5 point) Consider the following table, which is the output from running the optimal binary search tree algorithm for some seven-key input. In each entry, the value printed in the upper half of the cell is Tree$[i, j]$, the cost of the optimal binary search tree consisting of keys $i \ldots j$. The value printed in the lower half of the cell is the value of roots$[i, j]$.

   Suppose we have computed the following using the algorithm for the optimal binary search tree. However, we accidentally forgot to record Tree$[1, n]$. What is the missing value for that spot? It is the only one omitted in the following table.

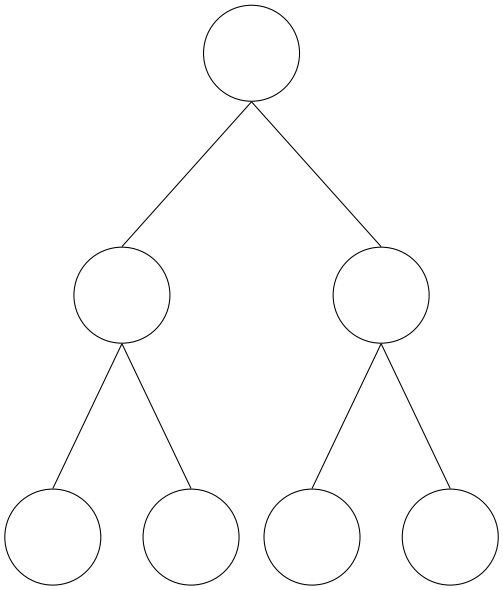
	k_1	k_2	k_3	k_4	k_5	k_6	k_7
k_1	0.17 1	0.41 1	0.8 2	1.29 3	1.87 4	2.17 4	 4
k_2		0.12 2	0.41 3	0.83 3	1.31 4	1.61 4	1.73 4
k_3			0.17 3	0.55 4	0.95 4	1.25 4	1.37 4
k_4				0.21 4	0.61 4	0.82 5	0.91 5
k_5					0.2 5	0.4 5	0.49 5
k_6						0.1 6	0.16 6
k_7							0.03 7

2. (1.5 point) Draw the root, the root's children, and the root's grandchildren for the optimal binary search tree for the input probabilities in the previous question.

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Write your answer to question 1 in this box.

Write the keys for question 2 into this tree. If a subtree is null, leave the entry blank.



You may use the rest of this page as scratch paper if you wish, but nothing outisde of the designated answer areas will be graded.

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3. (2 points) As a typical overworked college student, you occasionally pull all-nighters (refusing to sleep) to get more work done. Painful experience has taught you that the longer you stay awake, the less productive you are.

   Suppose there are n nights left in the quarter. For each night, you can either stay awake and work, or you can sleep. You have an array Score$[1 \dots n]$, where Score$[i]$ is the (always positive) number of points you will earn on day i *if you are awake and well rested.*

   However, staying awake for several days in a row has a price: each consecutive day you stay awake cuts the quality of your work in half. Thus, if you are awake on day i, and you most recently slept on day $i - k$, you will earn Score$[i]/2^{k-1}$ points on day i. You slept on day zero. On days you sleep, you earn zero points.

   For example, suppose $n = 6$ and Score $= [3, 7, 4, 3, 9, 1]$

   - If you work on all six days, you will earn $3 + \frac{7}{2} + \frac{4}{4} + \frac{3}{8} + \frac{9}{16} + \frac{1}{32} = 8.46875$ points
   - If you work only on days 1,3, and 5, you will earn $3 + 4 + 9$ points.
   - If you work only on days 2, 3, 5, and 6, you will earn $7 + \frac{4}{2} + 9 + \frac{1}{2} = 18.5$ points. This is optimal for the above input.

   Use dynamic programming to design an algorithm to compute the maximum number of points you can earn, given the array Score$[1 \dots n]$ as input. **Your professor encourages you to get a healthy amount of sleep and reminds you that, in real life, you can earn points and sleep on the same day.**

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Write your answer to question 3 on this page. It is possible that you will not use the entire page.

Scratch page. Nothing you write on this side of the page will be graded.

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